

Progress Test 1.A — Thermo-S1

1.A · System and state variables

Semester 1 · Academic Year 2026–2027

ANSWER KEY · MODEL ANSWERS AND MARKING SCHEME

Duration: 7 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. Unit conversion

(1 mark)

Convert the molar volume $V_m = 24.5 \text{ L mol}^{-1}$ into $\text{m}^3 \text{ mol}^{-1}$. Give the exact result, without rounding.

$1 \text{ L} = 1 \text{ dm}^3 = (10^{-1} \text{ m})^3 = 10^{-3} \text{ m}^3$	$V_m = 24.5 \times 10^{-3} \text{ m}^3 \text{ mol}^{-1}$
$V_m = 2.45 \times 10^{-2} \text{ m}^3 \text{ mol}^{-1}$	

Marking

①
0 1
☐ ☐

Marking scheme

① **1 pt** $2.45 \times 10^{-2} \text{ m}^3 \text{ mol}^{-1}$, with the unit written correctly.

Examiner's tip

A litre is a cubic decimetre: cube the conversion factor, $(10^{-1})^3 = 10^{-3}$. Every SI formula of this module ($pV = nRT$, ...) needs volumes in m^3 .

Exercise 2. Closed and isolated systems — definitions

(5 marks)

Define a closed system and an isolated system. For a gas enclosed in a cylinder, say what the walls must be like to prevent each kind of exchange with the surroundings. Is every closed system isolated?

Closed system: it exchanges no <i>matter</i> with its surroundings (n is constant). Isolated system: it exchanges neither <i>matter</i> nor <i>energy</i> with its surroundings. Walls needed for the gas to be isolated: • sealed walls, so that no matter is exchanged; • rigid walls and a fixed piston, so that no work is exchanged (pressure forces cannot move them); • thermally insulating walls, so that no heat is exchanged. An isolated system is necessarily closed, but the converse is false. For example, a sealed weather balloon is closed, yet it exchanges heat and work with the air around it.

Marking

①
0 1 2
☐ ☐ ☐

②
0 1 2
☐ ☐ ☐

③
0 1
☐ ☐

Marking scheme

- ① **2 pts** Closed: no exchange of matter (1 pt). Isolated: no matter and no energy (1 pt).
 ② **2 pts** Sealed, rigid or fixed, and insulating walls (1 pt if only two of the three).
 ③ **1 pt** Isolated $\Rightarrow$ closed, but closed $\nRightarrow$ isolated, with an example.

Exercise 3. Classifying quantities and systems

(4 marks)

Which of these quantities is intensive?

- ☐ the mass m ☐ the volume V
☒ the molar volume $V_m = V/n$ ☐ the energy E

Why: the quotient of two extensive quantities is intensive: doubling the system doubles both V and n , so V/n does not change.

A hot-air balloon, with its opening at the bottom, is:

- ☐ closed ☒ open ☐ isolated ☐ closed and isolated

Why: air enters and leaves through the opening, so the system exchanges matter (Fig. 1 of the notes).

The SI dimension of pressure is:

- ☒ $\text{ML}^{-1} \text{T}^{-2}$ ☐ MLT^{-2} ☐ $\text{ML}^2 \text{T}^{-2}$ ☐ $\text{ML}^{-2} \text{T}^{-2}$

Why: pressure is a force per unit area: $\text{MLT}^{-2}/\text{L}^2 = \text{ML}^{-1} \text{T}^{-2}$ (Tab. 1 of the notes).

 R is an intensive constant. Which relation is homogeneous in the extensive / intensive sense?

- ☐ $\frac{p}{T} = \frac{R}{V}$ ☐ $pV = RT$ ☒ $\frac{p}{T} = \frac{nR}{V}$ ☐ $p = nRT$

Why: p/T is intensive. So is nR/V (an extensive quantity divided by an extensive one). In the other options, one side doubles with the system and the other does not (Box 7).

Exercise 4. Composition of a solution

(4 marks)

A mass m of sodium chloride (molar mass M) is dissolved in water to obtain a volume V of solution. Compute the amount of substance n of solute, the mass concentration c_m and the molar concentration c . Check that $c_m = cM$.

$m = 5.85 \text{ g}$	$M = 58.5 \text{ g mol}^{-1}$	$V = 0.500 \text{ L}$
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$n = \frac{m}{M} = \frac{5.85 \text{ g}}{58.5 \text{ g mol}^{-1}}$	$n = 0.100 \text{ mol}$
$c_m = \frac{m}{V} = \frac{5.85 \text{ g}}{0.500 \text{ L}}$	$c_m = 11.7 \text{ g L}^{-1}$
$c = \frac{n}{V} = \frac{0.100 \text{ mol}}{0.500 \text{ L}}$	$c = 0.200 \text{ mol L}^{-1}$
Check: $cM = 0.200 \times 58.5 = 11.7 \text{ g L}^{-1} = c_m$. ✓ Both concentrations are intensive (quotients of extensive quantities).	

Marking

①	
0	1
②	
0	1
③	
0	1
④	
0	1

Marking scheme

- ① 1 pt $n = m/M = 0.100 \text{ mol}$.
- ② 1 pt $c_m = 11.7 \text{ g L}^{-1}$.
- ③ 1 pt $c = 0.200 \text{ mol L}^{-1}$.
- ④ 1 pt $c_m = cM$ checked.

Examiner's tip

Test 1, Ex. 4: write $n = m/M$ with letters first, then substitute *with units*. The units then cancel correctly by themselves: $\text{g}/(\text{g mol}^{-1}) = \text{mol}$.

Your score: / 14 → mark out of 20 = score $\times$ 20/14 = / 20